DIPLOMA PROGRAMME PRACTICE · SL · SET A

Biology / Paper 1A + 1B

Worked mark scheme · 55 marks · 90 minutes across 1A and 1B

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each.

PAPER 1A · QUESTION 1 [1]

Answer C Hydrogen bonds between water molecules contribute to surface tension. Reducing that tension reduces the supporting force.

PAPER 1A · QUESTION 2 [1]

Answer A The other 126 pairs are A–T, each containing exactly one thymine.

PAPER 1A · QUESTION 3 [1]

Answer D Convert 7.2 mm to 7200 µm. Image width divided by actual width is 7200/18=400.

PAPER 1A · QUESTION 4 [1]

Answer B The biological species concept relies on reproductive isolation. A difference in appearance, geography or allele frequency alone does not establish this.

PAPER 1A · QUESTION 5 [1]

Answer A A conserved arrangement of underlying structures supports common ancestry. Similarity alone cannot identify a direct ancestor.

PAPER 1A · QUESTION 6 [1]

Answer D Sampling unrelated individuals across the range reduces the loss of alleles through founder effects.

PAPER 1A · QUESTION 7 [1]

Answer B The active site recognizes particular linkages. Starch and cellulose differ in the arrangement of their glucose residues.

PAPER 1A · QUESTION 8 [1]

Answer D Activity depends on shape. Denaturation can disrupt interactions maintaining that shape while leaving the primary sequence intact.

PAPER 1A · QUESTION 9 [1]

Answer D Facilitated diffusion through an open channel is passive and dissipates a concentration gradient.

PAPER 1A · QUESTION 10 [1]

Answer B A secreted protein enters rough ER, is processed and sorted in Golgi, and reaches the membrane in a vesicle.

PAPER 1A · QUESTION 11 [1]

Answer C Area scales with length squared and volume with length cubed. Doubling length halves the ratio.

PAPER 1A · QUESTION 12 [1]

Answer C At fixed area and concentration difference the rate varies inversely with thickness.

PAPER 1A · QUESTION 13 [1]

Answer C The higher upstream pressure pushes the one-way valve open. Blood then moves down the pressure gradient from the chamber into the artery; the valve does not pump it.

PAPER 1A · QUESTION 14 [1]

Answer B Still, humid air near the stomata reduces the gradient that drives evaporation from the leaf.

PAPER 1A · QUESTION 15 [1]

Answer D The exclusion treatment shows the grazer can live higher on the shore. Biotic competition restricts its observed range.

PAPER 1A · QUESTION 16 [1]

Answer B Subtract non-enzymatic loss from the 1.8 mmol total and divide 1.5 mmol by 30 s.

PAPER 1A · QUESTION 17 [1]

Answer C Respiration transfers some substrate energy to ATP and some to the surroundings as heat.

PAPER 1A · QUESTION 18 [1]

Answer D Gross oxygen production equals net release plus respiratory consumption, 8+3.

PAPER 1A · QUESTION 19 [1]

Answer B An action potential propagates as successive regions of the membrane change electrical potential.

PAPER 1A · QUESTION 20 [1]

Answer A A stimulus is detected, conveyed to the central nervous system, integrated and followed by motor output to the iris muscles.

PAPER 1A · QUESTION 21 [1]

Answer C Memory cells generated during primary exposure allow faster clonal expansion and production of antigen-specific antibodies.

PAPER 1A · QUESTION 22 [1]

Answer B Marked proportion in the sample is 12/60. Equating that to 80/N gives N=400.

PAPER 1A · QUESTION 23 [1]

Answer D 54/600×100=9%. Ingested energy also leaves as wastes and is dissipated during respiration.

PAPER 1A · QUESTION 24 [1]

Answer A After two rounds there are four molecules per original, of which two retain one original labelled strand.

PAPER 1A · QUESTION 25 [1]

Answer A mRNA matches the coding strand in the same 5′ to 3′ direction, with U in place of T.

PAPER 1A · QUESTION 26 [1]

Answer B Degeneracy of the genetic code permits a synonymous substitution.

PAPER 1A · QUESTION 27 [1]

Answer C Replication produces two sister chromatids per chromosome. Chromosome number is counted by centromeres before separation.

PAPER 1A · QUESTION 28 [1]

Answer A Opposite changes in tissue mass are consistent with opposite net water movement by osmosis.

PAPER 1A · QUESTION 29 [1]

Answer A Each offspring has probability ½ of inheriting the recessive allele from the heterozygote. Independent events give ½×½.

PAPER 1A · QUESTION 30 [1]

Answer A Evolution by selection requires heritable differences associated with differential reproductive success.

PAPER 1B · QUESTION 1 [6]

Measuring a stained epidermis

(a) Estimate the bracketed length of cell A. Show the ratio you used. [2]

1 The bracket measures approximately 3.3 scale bars (allow 3.0–3.6).

1 Actual length is about 165 µm (accept 150–180 µm with consistent working).

(b) Calculate the magnification on the separate display. [2]

1 12.0 mm=12000 µm.

1 Magnification=12000/50=240×.

(c) Give one visible feature supporting a plant origin. Explain why the dark boundary alone does not distinguish the plasma membrane from the cell wall at this resolution. [2]

1 Closely fitting cells with clear rigid walls or a regular angular outline support plant tissue.

1 The membrane lies very close to the wall and is too thin to be resolved separately by ordinary light microscopy.

PAPER 1B · QUESTION 2 [6]

Ventilation and exchange

(a) Calculate the volume reaching exchange surfaces each minute at rest and during exercise. [2]

1 At rest: $12\times(0.50-0.15)=4.20$ dm³ min⁻¹.

1 During exercise: $24\times(1.10-0.15)=22.8$ dm³ min⁻¹.

(b) Under the model, compare slow breathing at 10 breaths of 0.60 dm³ per minute with the resting pattern. Explain the difference despite equal total ventilation. [2]

1 Slow breathing delivers $10\times0.45=4.50$ dm³ min⁻¹, higher than 4.20.

1 The same total 6.0 dm³ is moved, but less is spent filling the conducting airways over fewer breaths.

(c) Explain why more ventilation need not produce a proportional increase in oxygen uptake by the body. [2]

1 Gas exchange also requires a partial-pressure or concentration gradient and adequate perfusion.

1 If tissues use little additional oxygen or blood flow limits transfer, extra inspired air need not yield a matching uptake.

PAPER 1B · QUESTION 3 [6]

Separating enzyme activity from background change

(a) Calculate the corrected mean absorbance change per second at 35 °C. [2]

1 Corrected change=0.260−0.060=0.200.

1 Rate=0.200/40=0.00500 absorbance units s^{-1}.

(b) Compare the corrected rates at 25 °C and 45 °C with evidence. [2]

1 Corrected changes are 0.160 and 0.140 over the same time (rates 0.00400 and 0.00350 s^{-1}).

1 The 45 °C corrected rate is lower despite its larger uncorrected change.

(c) Evaluate the claim that 35 °C is the enzyme's exact optimum. [2]

1 It gives the highest corrected activity among the tested temperatures.

1 Closer temperatures and independent repeats with uncertainty are needed to locate the optimum more precisely.

PAPER 1B · QUESTION 4 [7]

An isotonic estimate from plant tissue

(a) Estimate the concentration for zero net mass change by linear interpolation. [2]

1 The crossing is between 0.20 and 0.30 mol dm^{-3}.

1 It is 3/5 of the interval above 0.20, giving about 0.26 mol dm^{-3}.

(b) A matched cylinder initially weighs 2.50 g. Predict its final mass at 0.40 mol dm^{-3} and explain the sign of the change. [2]

1 Final mass=2.50×0.93=2.325 g (about 2.33 g).

1 Water leaves cells by osmosis because the external solution has lower water potential.

(c) Explain one reason to blot samples consistently and one limitation of using all cylinders from a single tuber. [2]

1 Unremoved surface liquid adds variable mass unrelated to water movement in the tissue.

1 One tuber does not estimate between-plant variation, limiting generalization.

(d) State what continues at the membrane at the zero crossing. [1]

1 Water molecules continue crossing both ways at equal average rates.